Maximum SDP distances in size-n KMSW cells

Abstract

Let $M_{0,n}$ be the size-n KMSW cell obtained from n independent increments sampled uniformly from $\{(1,-1),(-1,0),(0,1)\}$. We prove that, for every $\varepsilon > 0$, the maximum SDP distance over pairs of vertices which can be joined by a directed path in $M_{0,n}$ is at most $n^{3/8+\varepsilon}$ with probability tending to one as $n \to \infty$. The proof uses the shortest-path Busemann function of Borga–Gwynne. The main new step is an exact reversal of a finite KMSW encoding with one marked boundary vertex. After the reversal, the marked distance is a negative sum of independent Busemann increments along the right boundary. The one-sided bound on these increments gives a stretched-exponential estimate. The future-map description of the UIQBOT then transfers this estimate to every reachable ordered pair in the finite cell. Together with the lower bound of Borga–Gwynne, this gives $m_n = n^{3/8+o(1)}$ for any median m_n of the maximum distance.

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1 Introduction and main result

We use the notation of Borga–Gwynne [BG]. Let $\mathcal{Z} = (\mathcal{L}, \mathcal{R})$ be a walk on $\mathbb{Z}^2$, started from $(0,0)$, whose increments are independent and sampled uniformly from

$$(1,-1), \qquad (-1,0), \qquad (0,1). \tag{1.1}$$

Applying the KMSW procedure of Kenyon–Miller–Sheffield–Wilson [KMSW19] to $\mathcal{Z}|_{[0,n]}$ gives a finite directed map $M_{0,n}$, possibly with missing boundary edges. We denote its time-j active edge by $\lambda_j = (\lambda_j^-, \lambda_j^+)$.

For vertices x, y of a directed map M, let $\mathrm{SDP}_M(x,y)$ be the minimum number of edges in a directed path from x to y, with value ∞ if there is no such path. Set

$$D_n := \max\left\{\mathrm{SDP}_{M_{0,n}}(x,y) : x,y \in \mathcal{V}(M_{0,n}),\ \mathrm{SDP}_{M_{0,n}}(x,y) < \infty\right\}. \tag{1.2}$$

The following is the main result.

Theorem 1.1. *For every $\varepsilon > 0$ and every $p > 0$, there is a constant $C = C(\varepsilon, p) > 0$ such that*

$$\mathbb{P}\left[D_n \geq n^{3/8+\varepsilon}\right] \leq Cn^{-p}, \qquad n \in \mathbb{N}. \tag{1.3}$$

In particular,

$$\lim_{n\to\infty} \mathbb{P}\left[D_n \geq n^{3/8+\varepsilon}\right] = 0. \tag{1.4}$$

Let m_n be any median of D_n, i.e., $\mathbb{P}[D_n \leq m_n] \geq 1/2$ and $\mathbb{P}[D_n \geq m_n] \geq 1/2$. The third assertion of [BG, Theorem 1.13] gives the matching lower bound for D_n. Consequently, Theorem 1.1 has the following immediate consequence.

Corollary 1.2. *For any choice of medians m_n,*

$$\lim_{n\to\infty} \frac{\log m_n}{\log n} = \frac{3}{8}. \tag{1.5}$$

Equivalently, $m_n = n^{3/8+o(1)}$.

We now explain the proof. Let $\widehat{M}$ be the UIQBOT and let o be its source. If $\widehat{\lambda}_t$ is the time-t active edge, we first estimate

$$a_t := \mathrm{SDP}_{\widehat{M}_{0,t}}(o, \widehat{\lambda}_t^-). \tag{1.6}$$

We extend the first t increments to an encoding of a finite bipolar-oriented triangulation in such a way that $\widehat{\lambda}_t^-$ remains on its upper-left boundary. We then reverse every edge orientation and reverse the order of the KMSW increments. A further infinite walk segment makes the old source a cut vertex. The distance in (1.6) is then a Busemann difference between two vertices of the right boundary.

In the SDP case, the right-boundary Busemann increments are independent, have mean zero, are bounded below by -1, and have positive tail of order $x^{-4/3}$ [BG, Theorems 1.7 and 1.8]. It follows that a negative sum over k consecutive increments has scale $k^{3/4}$ and has a stretched-exponential upper tail. The number of relevant boundary edges in our finite map is of order $t^{1/2+o(1)}$, which gives the exponent $(1/2)(3/4) = 3/8$.

Finally, the future-map result of [BG] says that the submap induced by the vertices reachable from the initial vertex of an active edge has the UIQBOT law. Applying this result at the first edge of a directed path in $M_{0,n}$ reduces every finite SDP in $M_{0,n}$ to a source distance in one of at most $n+1$ maps, each having the UIQBOT law. The estimate for (1.6) is strong enough to take a union bound over all of them.

2 KMSW and Busemann inputs

This section records the facts from [BG, KMSW19] which we use. We state them only in the triangulation setting.

2.1 The UIQBOT encoding walk

Let $\widehat{\mathcal{Z}} = (\widehat{\mathcal{L}}, \widehat{\mathcal{R}})$ be the walk with increments as in (1.1), with its first coordinate conditioned to stay non-negative. If the walk starts from $(x, y) \in \mathbb{N}_0 \times \mathbb{Z}$, its law on the first m increments is obtained by weighting the unconditioned walk law by

$$\frac{\mathcal{L}(m)+1}{x+1}\mathbf{1}_{\{\mathcal{L}(j)\geq 0,\ 0\leq j\leq m\}}. \tag{2.1}$$

The infinite map obtained by applying the KMSW procedure to $\widehat{\mathcal{Z}}$ is the UIQBOT $\widehat{M}$ [BG, Definition 4.1]. Throughout, $\widehat{\mathbb{P}} = \widehat{\mathbb{P}}_{(0,0)}$ denotes the law of this conditioned walk.

We will use the following consequence of (2.1). For each $m \in \mathbb{N}$ and $U \geq \sqrt{m}$,

$$\widehat{\mathbb{P}}\left[\max_{0\leq j\leq m}\left(|\widehat{\mathcal{L}}(j)| + |\widehat{\mathcal{R}}(j)|\right) > U\right] \leq Cm^C \exp\left(-c\frac{U^2}{m}\right). \tag{2.2}$$

Indeed, under the unconditioned law the corresponding estimate follows from the bounded increment maximal inequality. Formula (2.1) and Cauchy–Schwarz introduce only a power of m.

We use the pathwise form of the future-map construction. Let $M_{0,\infty}$ be obtained from the time-0 future of the UIBOT encoding walk, let $x_0 = \lambda_0^-$, and let $\widehat{M}_{0,\infty}$ be the submap induced by the vertices reachable from x_0. Write

$$0 = S_0 < S_1 < S_2 < \cdots \tag{2.3}$$

for the times whose active edges belong to $\widehat{M}_{0,\infty}$. Define $\widehat{\mathcal{Z}}(0) = 0$ and

$$\widehat{\mathcal{Z}}(k) - \widehat{\mathcal{Z}}(k-1) := \mathcal{Z}(S_k) - \mathcal{Z}(S_k - 1), \qquad k \geq 1. \tag{2.4}$$

By [BG, Lemmas 5.13 and 5.14], applying the KMSW procedure to $\widehat{\mathcal{Z}}$ gives $\widehat{M}_{0,\infty}$, with time-k active edge equal to λ_{S_k}, and $\widehat{\mathcal{Z}}$ has the conditioned law described in (2.1). We henceforth use this coupling whenever the UIQBOT and $M_{0,\infty}$ occur together.

The right boundary of $\widehat{M}_{0,\infty}$ is the rightmost directed path in $M_{0,\infty}$ from x_0 to infinity; it is the non-missing part of the boundary of $M_{0,\infty}$, as in the setup preceding [BG, Proposition 5.1]. Thus it is the path with vertices $x_0, x_1, x_2, \ldots$ used to define the positive-index Busemann increments. Moreover, every directed path in $M_{0,\infty}$ started from a vertex of $\widehat{M}_{0,\infty}$ stays in $\widehat{M}_{0,\infty}$: one concatenates it with a directed path from x_0 to its initial vertex.

2.2 Comparison of two KMSW explorations

We record a deterministic consequence of the KMSW procedure which will be used twice below.

Lemma 2.1. *Let $\mathfrak{Z} = (\mathfrak{L}, \mathfrak{R}) : [0, n] \cap \mathbb{Z} \to \mathbb{Z}^2$ be a KMSW walk, and let $\lambda_i = (\lambda_i^-, \lambda_i^+)$ be its active edges. For $0 \leq i \leq j \leq n$,*

$$\lambda_i^- = \lambda_j^- \iff \mathfrak{L}(i) = \mathfrak{L}(j) = \min_{i\leq s\leq j} \mathfrak{L}(s), \tag{2.5}$$

$$\lambda_i^+ = \lambda_j^+ \iff \mathfrak{R}(i) = \mathfrak{R}(j) = \min_{i\leq s\leq j} \mathfrak{R}(s). \tag{2.6}$$

For $i < j$, one has $\lambda_i^+ = \lambda_j^-$ if and only if there is $q \in (i, j] \cap \mathbb{Z}$ such that

$$\mathfrak{Z}(q) - \mathfrak{Z}(q-1) = (1, -1), \tag{2.7}$$

$$\mathfrak{R}(i) = \mathfrak{R}(q-1) = \min_{i\leq s\leq q-1} \mathfrak{R}(s), \tag{2.8}$$

$$\mathfrak{L}(q) = \mathfrak{L}(j) = \min_{q\leq s\leq j} \mathfrak{L}(s). \tag{2.9}$$

Moreover, $\lambda_i^+ \neq \lambda_j^-$ whenever $i \geq j$. Consequently, active times increase strictly along every directed path.

Now fix $0 \leq a < b \leq n$ and apply the KMSW procedure to the shifted restriction

$$\mathfrak{Z}^{a,b}(s) := \mathfrak{Z}(a+s) - \mathfrak{Z}(a), \qquad 0 \leq s \leq b-a. \tag{2.10}$$

Index its active edges by the original times and denote them by $\lambda_i^{a,b}$, $a \leq i \leq b$. Then, for every $i, j \in [a,b] \cap \mathbb{Z}$ and $\sigma, \tau \in \{-,+\}$,

$$(\lambda_i^{a,b})^\sigma = (\lambda_j^{a,b})^\tau \quad \Longleftrightarrow \quad \lambda_i^\sigma = \lambda_j^\tau. \tag{2.11}$$

In particular, the correspondence $\lambda_i^{a,b} \mapsto \lambda_i$ preserves directed edge incidences.

Proof. While $\mathfrak{L}$ stays at least $\mathfrak{L}(i)$, the upper-left boundary distance between λ_i^- and λ_j^- is $\mathfrak{L}(j) - \mathfrak{L}(i)$. When $\mathfrak{L}$ first goes strictly below $\mathfrak{L}(i)$, the vertex λ_i^- is passed by the upper-left boundary and is never again the initial vertex of an active edge. This proves (2.5). For completeness, this boundary-distance identity follows inductively from the three KMSW steps: a $(1,-1)$ step moves the active initial vertex one edge forward on the upper-left boundary, a $(-1,0)$ step moves it one edge backward, and a $(0,1)$ step leaves it unchanged. The backward move which would make the displayed distance negative removes λ_i^- from that boundary, after which the KMSW procedure never restores it. Exchanging initial with terminal, upper-left with lower-right, and $\mathfrak{L}$ with $\mathfrak{R}$ gives (2.6) in the same manner.

The only KMSW operation which makes the terminal vertex of the preceding active edge the initial vertex of the new active edge is a $(1,-1)$ increment. At such a time q,

$$\lambda_{q-1}^+ = \lambda_q^-. \tag{2.12}$$

Equation (2.6) says that $\lambda_i^+ = \lambda_{q-1}^+$ exactly under (2.8); equation (2.5) says that $\lambda_q^- = \lambda_j^-$ exactly under (2.9). This proves the mixed endpoint criterion.

At each vertex, every occurrence as the terminal vertex of an active edge precedes every occurrence as the initial vertex of an active edge. We verify this inductively from Definition 3.3 of [BG]. It holds for the two endpoints of the initial active edge. A $(-1,0)$ step continues the consecutive collection of active edges terminating at the same vertex; its other endpoint is on the upper-left boundary and is the initial vertex of the new active edge. A $(0,1)$ step continues the consecutive collection of active edges starting at the same vertex and introduces a new terminal endpoint. A $(1,-1)$ step identifies the terminal endpoint of the preceding active edge with the initial endpoint of the new active edge, whether the new edge is created or a missing upper-right boundary edge is activated. This is the unique transition from terminal occurrences to initial occurrences. None of the three operations permits the reverse transition. The induction proves the assertion, and hence $\lambda_i^+ \neq \lambda_j^-$ for $i \geq j$. If λ_i is followed by λ_j in a directed path, then $\lambda_i^+ = \lambda_j^-$, so $i < j$.

Finally, all of the conditions (2.5)–(2.9) depend only on the increments and coordinate differences between the two relevant times. These data are unchanged by (2.10). Applying the same criteria to the two KMSW explorations proves (2.11) for equal endpoint types and for a terminal endpoint followed by an initial endpoint. The remaining mixed case follows by exchanging i and j and using the ordering of terminal and initial occurrences. This proves the claim for all four choices of σ and τ. □

2.3 The SDP Busemann increments

Let $x_0, x_1, x_2, \dots$ be the vertices on the right boundary of $M_{0,\infty}$, ordered so that (x_{k-1}, x_k) is directed from x_{k-1} to x_k. Let $\mathcal{X}$ be the SDP Busemann function of [BG], normalized so that $\mathcal{X}(0) = 0$, and set

$$Y_k := \mathcal{X}(k) - \mathcal{X}(k-1), \qquad k \geq 1. \tag{2.13}$$

By [BG, Theorems 1.7 and 1.8], the random variables Y_k are independent and identically distributed, and

$$Y_k \geq -1, \qquad \mathbb{E}[Y_k] = 0, \qquad \mathbb{P}[Y_k > u] \leq Cu^{-4/3}, \quad u \geq 1. \tag{2.14}$$

The next estimate is the form in which we will use these properties.

Lemma 2.2. *There are constants $c, C > 0$ such that, for $K \in \mathbb{N}$ and $h \geq K^{3/4}$,*

$$\mathbb{P}\left[\max_{0 \leq i < j \leq K} \left(-\sum_{k=i+1}^{j} Y_k\right) \geq h\right] \leq CK \exp\left(-c\frac{h^4}{K^3}\right). \tag{2.15}$$

If $h > K$, the probability on the left side is zero.

Proof. For sufficiently small $\lambda > 0$, (2.14) and $\mathbb{E}[Y_1] = 0$ give

$$\mathbb{E}[e^{-\lambda Y_1}] = 1 + \mathbb{E}[e^{-\lambda Y_1} - 1 + \lambda Y_1] \leq \exp(C\lambda^{4/3}). \tag{2.16}$$

To justify the last inequality, put $g_\lambda(y) = e^{-\lambda y} - 1 + \lambda y$. Since $Y_1 \geq -1$,

$$\begin{aligned} &g_\lambda(Y_1) \leq C\lambda^2, && -1 \leq Y_1 \leq 0, \\ &g_\lambda(Y_1) \leq C\lambda^2 Y_1^2, && 0 \leq Y_1 \leq \lambda^{-1}, \\ &g_\lambda(Y_1) \leq \lambda Y_1, && Y_1 > \lambda^{-1}. \end{aligned}$$

The tail bound in (2.14) gives

$$\mathbb{E}\left[Y_1^2 \mathbf{1}_{\{0 \leq Y_1 \leq \lambda^{-1}\}}\right] \leq C\lambda^{-2/3}, \qquad \mathbb{E}\left[Y_1 \mathbf{1}_{\{Y_1 > \lambda^{-1}\}}\right] \leq C\lambda^{1/3}.$$

Consequently, $\mathbb{E}[g_\lambda(Y_1)] \leq C\lambda^{4/3}$, which proves (2.16).

Fix $i \in [0, K-1] \cap \mathbb{Z}$. The exponential maximal inequality applied to (2.16) gives

$$\mathbb{P}\left[\max_{i < j \leq K} -\sum_{k=i+1}^{j} Y_k \geq h\right] \leq \exp\{-\lambda h + CK\lambda^{4/3}\}. \tag{2.17}$$

For $K^{3/4} \leq h \leq K$, choose $\lambda = c_0(h/K)^3$, with $c_0 > 0$ sufficiently small. This makes the right side of (2.17) at most $\exp(-ch^4/K^3)$. A union bound over i proves (2.15). If $h > K$, the event is impossible since $-Y_k \leq 1$. □

3 Reversal of a finite KMSW encoding

In this section we compare the distance in (1.6) to the Busemann increments in (2.13). We first record a random walk estimate which permits us to extend a finite segment of the UIQBOT encoding walk so that the KMSW procedure produces a finite bipolar-oriented triangulation.

3.1 A random walk estimate

Lemma 3.1. *Fix $A > 1$. There are $c = c(A) > 0$ and $m_0 = m_0(A) \in \mathbb{N}$ with the following property. Let $m \geq m_0$, and let $d, d' \in [A^{-1}\sqrt{m}, A\sqrt{m}] \cap \mathbb{Z}$. Assume that $(d', 0)$ can be reached from $(0, d)$ in m steps of the walk in (1.1). Then*

$$\mathbb{P}_{(0,d)}\left[\mathcal{Z}[0,m] \subset \mathbb{N}_0^2,\ \mathcal{Z}(m) = (d', 0)\right] \geq cm^{-2}. \tag{3.1}$$

Proof. Put $C_+ = \mathbb{N}_0^2$ and, for $z, z' \in C_+$, let

$$p_k^+(z, z') := \mathbb{P}_z\left[\mathcal{Z}[0,k] \subset C_+,\ \mathcal{Z}(k) = z'\right]. \tag{3.2}$$

We use the three-interval argument from the proof of [BG, Lemma 5.17]. Choose $m_1, m_2, m_3 \in [m/4, m/2] \cap \mathbb{N}$ such that $m_1 + m_2 + m_3 = m$ and $m_2 \in 3\mathbb{Z}$. Let K be a square compactly contained in $(0, \infty)^2$, symmetric under exchange of the two coordinates, and chosen large enough in terms of A.

The one-dimensional ballot estimate and the half-plane conditioned invariance principle [DW20, Theorem 1] give, uniformly in the present range of d,

$$\sum_{u \in \sqrt{m}K \cap \mathbb{Z}^2} p_{m_1}^+((0,d), u) \geq c_1 m^{-1/2}. \tag{3.3}$$

Indeed, requiring the first coordinate to remain non-negative has probability at least $cm^{-1/2}$. Under this conditioning, the rescaled walk converges to the correlated Brownian motion conditioned to remain in a half-plane. Since $d/\sqrt{m}$ ranges over a compact subset of $(0, \infty)$, that limiting process has a uniformly positive probability that its second coordinate remains positive and its endpoint lies in K. This is exactly the argument for the first interval in the proof of [BG, Lemma 5.17]. For the terminal portion, reverse the order of its increments and exchange the two coordinates. This preserves their law and preserves C_+. Consequently, the same estimate gives

$$\sum_{v \in \sqrt{m}K \cap \mathbb{Z}^2} p_{m_3}^+(v, (d', 0)) \geq c_2 m^{-1/2}. \tag{3.4}$$

Here and below, each sum is restricted to the period-three class which is reachable in the indicated number of steps.

For every pair u, v occurring in these sums, the lattice local limit theorem [Spi76, Section 7, P9] and the invariance principle for endpoint-conditioned sums [Lig70, Theorem 3] imply

$$p_{m_2}^+(u, v) \geq c_3 m^{-1}. \tag{3.5}$$

Indeed, after rescaling by $m^{-1/2}$ the two endpoints stay in the compact set K. The local limit estimate gives the factor m^{-1}. A Brownian bridge between two points of K has a positive probability to remain in the open quadrant; this probability is bounded below uniformly over the two endpoints by compactness. The bridge invariance principle gives the same uniform lower bound for the walk bridge when m is large. The reachability assumption in the lemma, together with $m_2 \in 3\mathbb{Z}$, makes the two endpoint classes compatible. More explicitly, the period class of a point z is determined by $z_1 - z_2$ modulo 3; feasibility of the first and last intervals and $m + d + d' \in 3\mathbb{Z}$ imply $(v_1 - v_2) - (u_1 - u_2) \equiv -m_2 \pmod 3$. Therefore, by the Chapman–Kolmogorov identity,

$$\begin{aligned} p_m^+((0,d), (d', 0)) &\geq \sum_{u,v \in \sqrt{m}K \cap \mathbb{Z}^2} p_{m_1}^+((0,d), u) p_{m_2}^+(u, v) p_{m_3}^+(v, (d', 0)) \\ &\geq cm^{-2}. \end{aligned} \tag{3.6}$$

This is (3.1). This is the argument of [BG, Lemma 5.17], except that the final segment is treated, after time reversal and coordinate exchange, in the same way as the initial segment. In [BG, Lemma 5.17], the final segment starts from the tip of the quadrant and has probability of order $m^{-3/2}$; this accounts for the exponent m^{-3} there instead of m^{-2} here. □

For $t \in \mathbb{N}$ and $U \geq \sqrt{t}$, define

$$E_t(U) := \left\{ \max_{0 \leq s \leq t} (|\widehat{\mathcal{L}}(s)| + |\widehat{\mathcal{R}}(s)|) \leq U \right\}. \tag{3.7}$$

Lemma 3.2. *There are constants $c, C > 0$ and $U_0 \in \mathbb{N}$ such that the following is true. Let $U \in \mathbb{N}$ with $U \geq U_0 \vee \sqrt{t}$. On $E_t(U)$, conditionally on $\widehat{\mathcal{Z}}|_{[0,t]}$, with probability at least cU^{-4}, the integer*

$$N := t + 12U^2 + 1 \tag{3.8}$$

satisfies

$$\widehat{\mathcal{L}}(s) \geq \widehat{\mathcal{L}}(t), \qquad t \leq s \leq N-1, \tag{3.9}$$

$$\widehat{\mathcal{R}}(N) < \min_{0 \leq s < N} \widehat{\mathcal{R}}(s), \tag{3.10}$$

$$0 < \widehat{\mathcal{L}}(N) \leq CU, \qquad 0 < -\widehat{\mathcal{R}}(N) \leq CU. \tag{3.11}$$

Proof. Write $\underline{R}_t = \min_{0 \leq s \leq t} \widehat{\mathcal{R}}(s)$. Choose integers as follows. Set

$$R_* := \underline{R}_t - 3U, \qquad d := \widehat{\mathcal{R}}(t) - (R_* + 1). \tag{3.12}$$

Let $\xi \in \{0, 1, 2\}$ be chosen so that

$$12U^2 + d + 3U + \xi \in 3\mathbb{Z}, \qquad \ell := \widehat{\mathcal{L}}(t) + 3U + \xi + 1. \tag{3.13}$$

Before the last increment, require the walk to stay in

$$[\widehat{\mathcal{L}}(t), \infty) \times [R_* + 1, \infty) \tag{3.14}$$

and to end at $(\ell - 1, R_* + 1)$. Require the last increment to be $(1, -1)$, so that $\widehat{\mathcal{Z}}(N) = (\ell, R_*)$. After translating the southwest corner, the portion before the final increment goes from $(0, d)$ to $(d', 0)$ in $12U^2$ steps, where

$$d' = 3U + \xi, \qquad 3U - 1 \leq d \leq 5U - 1. \tag{3.15}$$

The congruence in (3.13) is exactly the period-three condition for this endpoint. Since $12U^2$ is much larger than $d + d'$, the endpoint is reachable when $U \geq U_0$. Lemma 3.1 and the prescribed final increment therefore give probability at least cU^{-4} under the unconditioned walk law.

The event in (3.14) implies (3.9)–(3.11), after increasing the universal constant in (3.11). Moreover, the first coordinate at the end is at least its value at time t. By the weighting formula (2.1), passing from the unconditioned walk to $\widehat{\mathcal{Z}}$ does not decrease the probability of this event. This proves the lemma. □

3.2 Reversing the KMSW encoding walk

Fix a possible realization $\mathfrak{z} = (\mathfrak{l}, \mathfrak{r}) : [0, N] \cap \mathbb{Z} \to \mathbb{Z}^2$ of the UIQBOT encoding walk on the event in Lemma 3.2, and write

$$\mathfrak{z}(N) = (\ell, -r). \tag{3.16}$$

Condition (3.10) implies that the final increment is $(1, -1)$. The restriction $\mathfrak{z}|_{[0,N-1]}$ ends at $(\ell - 1, -r + 1)$ and satisfies

$$\mathfrak{l}(j) \geq 0, \qquad \mathfrak{r}(j) \geq \mathfrak{r}(N-1) = -r + 1, \qquad 0 \leq j \leq N - 1. \tag{3.17}$$

The KMSW bijection therefore shows that this restriction encodes a bipolar-oriented triangulation $\mathfrak{m}$ with N edges and left and right boundary lengths ℓ and r, respectively. In particular, $\mathfrak{m}$ has no missing edges.

Define a second walk path $\mathfrak{z}^\dagger = (\mathfrak{l}^\dagger, \mathfrak{r}^\dagger)$ by

$$\begin{aligned} \mathfrak{l}^\dagger(j) &:= \mathfrak{r}(N-1-j) - \mathfrak{r}(N-1), \\ \mathfrak{r}^\dagger(j) &:= \mathfrak{l}(N-1-j) - \mathfrak{l}(N-1), \qquad\qquad 0 \leq j \leq N-1, \\ \mathfrak{z}^\dagger(N) &:= \mathfrak{z}^\dagger(N-1) + (1, -1). \end{aligned} \tag{3.18}$$

Let $\mathfrak{m}^\dagger$ be obtained from $\mathfrak{m}$ by reversing every edge orientation, with the same outer face and cyclic ordering of edges, and root it at the orientation reversal of λ_{N-1}.

Lemma 3.3. *The restriction $\mathfrak{z}^\dagger|_{[0,N-1]}$ is the KMSW encoding walk of $\mathfrak{m}^\dagger$. The old left boundary is the right boundary of $\mathfrak{m}^\dagger$, and its left and right boundary lengths are r and ℓ. If $\lambda_0^\dagger, \dots, \lambda_{N-1}^\dagger$ are the active edges for $\mathfrak{m}^\dagger$, then*

$$\lambda_j^\dagger = (\lambda_{N-1-j}^+, \lambda_{N-1-j}^-), \qquad 0 \leq j \leq N-1. \tag{3.19}$$

Furthermore,

$$\mathfrak{z}^\dagger(N-1) = (r-1, -\ell+1), \qquad \mathfrak{z}^\dagger(N) = (r, -\ell), \tag{3.20}$$

the first coordinate of $\mathfrak{z}^\dagger$ is non-negative, and the time-N active edge has initial vertex equal to the source o of $\mathfrak{m}$.

Proof. Recall the upper-left and lower-right trees in the inverse KMSW procedure of [KMSW19, Section 2.1]. Reversing every edge orientation exchanges these two trees and reverses the direction of the Peano path between them. The root of the reversed map is therefore the orientation reversal of the last edge in the original Peano order. Since the clockwise contour exploration of the lower-right tree of $\mathfrak{m}$ visits $\lambda_0, \dots, \lambda_{N-1}$ in this order, the counterclockwise contour exploration of the upper-left tree of $\mathfrak{m}^\dagger$ visits the orientation reversals of $\lambda_{N-1}, \dots, \lambda_0$ in this order. The inverse KMSW procedure therefore gives (3.19).

It remains to identify the increments. A $(1, -1)$ increment, read in the opposite order with every edge orientation reversed, is again a $(1, -1)$ increment. A $(-1, 0)$ increment corresponds to two successively explored sides of a triangular face with a common terminal vertex; after reversal they have a common initial vertex, which gives a $(0, 1)$ increment. The same argument with initial and terminal vertices exchanged sends a $(0, 1)$ increment to a $(-1, 0)$ increment. These are exactly the increments in (3.18). The uniqueness in the KMSW bijection gives the asserted planar map, including all face incidences and parallel edges. It also exchanges the two boundary paths.

Equation (3.18) and (3.10) give

$$\mathfrak{l}^\dagger(j) = \mathfrak{r}(N-1-j) - \mathfrak{r}(N-1) \geq 0, \qquad 0 \leq j \leq N-1. \tag{3.21}$$

The endpoint identities in (3.20) follow by direct summation. In particular, the final $(1,-1)$ increment is a strict new minimum of the second coordinate. Finally, $\lambda^\dagger_{N-1} = (\lambda^+_0, \lambda^-_0)$ has terminal vertex $o = \lambda^-_0$. Since $\mathfrak{m}^\dagger$ has no missing edges, the final $(1,-1)$ increment creates an active edge whose initial vertex is o. □

Let $\mathcal{Z}^\dagger = (\mathcal{L}^\dagger, \mathcal{R}^\dagger)$ be a UIQBOT encoding walk, and let $B^\dagger_N$ be the event that $\mathcal{L}^\dagger(j) \geq r$ for every $j \geq N$. Conditional on $\mathcal{Z}^\dagger|_{[0,N]} = \mathfrak{z}^\dagger$, the gambler's ruin formula in [BG, Lemma 4.10] gives

$$\widehat{\mathbb{P}}\left[B^\dagger_N \mid \mathcal{Z}^\dagger|_{[0,N]} = \mathfrak{z}^\dagger\right] = \frac{1}{r+1}. \tag{3.22}$$

The strict minimum in (3.20) and the event $B^\dagger_N$ are exactly the two walk conditions in the cut-vertex criterion of [BG, Lemma 4.8]. Hence the old source o is a cut vertex on $B^\dagger_N$.

We need the following more precise consequence of these two conditions.

Lemma 3.4. *On $B^\dagger_N$, the submap $\mathfrak{m}^\dagger$ and the submap explored from time N onward intersect only at o. Consequently, every directed path from a vertex of $\mathfrak{m}^\dagger \setminus \{o\}$ to a vertex explored after time N passes through o.*

Proof. Let $i < N \leq j$. We apply Lemma 2.1 to the UIQBOT encoding walk $\mathcal{Z}^\dagger$. If $(\lambda^\dagger_i)^- = (\lambda^\dagger_j)^-$, then (2.5) and the inequalities $\mathcal{L}^\dagger(s) \geq \mathcal{L}^\dagger(N)$ for $s \geq N$ show that their common first-coordinate minimum is $\mathcal{L}^\dagger(N)$. A second application of (2.5), with one of the times equal to N, shows that the common vertex is $(\lambda^\dagger_N)^- = o$. Equality of the two terminal vertices is impossible by (2.6), since $\mathcal{R}^\dagger(N) < \mathcal{R}^\dagger(i)$.

An initial vertex at a time less than N cannot equal a terminal vertex at a time at least N, by the ordering of terminal and initial occurrences in Lemma 2.1. Finally, suppose that $(\lambda^\dagger_i)^+ = (\lambda^\dagger_j)^-$. Let q be as in (2.7)–(2.9). If $q > N$, then (2.8) is contradicted by the strict minimum of the second coordinate at time N. If $q \leq N$, then (2.9) and $B^\dagger_N$ imply $\mathcal{L}^\dagger(q) = \mathcal{L}^\dagger(N)$, so the common vertex is again $(\lambda^\dagger_N)^- = o$ by (2.5).

Every edge of $\mathfrak{m}^\dagger$ has active time less than N, whereas every edge in the remaining KMSW exploration has active time at least N. The endpoint comparisons above therefore prove the first assertion. The second follows immediately. □

The preceding cut statement does not by itself identify the boundary of the infinite map. The following boundary statement supplies this identification.

Lemma 3.5. *Let $z_0, z_1, \ldots, z_\ell$ be the vertices on the right boundary of $\mathfrak{m}^\dagger$, in directed order, so that $z_\ell = o$. In the infinite KMSW map encoded by $\mathcal{Z}^\dagger$, each edge (z_{q-1}, z_q), $1 \leq q \leq \ell$, remains on the right boundary. Under the coupling (2.3)–(2.4), this boundary is the path with vertices $x_0, x_1, x_2, \ldots$ in $M_{0,\infty}$. Hence $z_q = x_q$ for $0 \leq q \leq \ell$. On $B^\dagger_N$, every x_k with $k > \ell$ belongs to the part explored after time N.*

Proof. Since $\mathfrak{m}^\dagger$ has no missing edges, its right boundary is its lower-right boundary. Under the KMSW procedure, an edge added to the lower-right boundary is never subsequently removed from that boundary; equivalently, these edges are indexed by the successive running minima of the second coordinate. This is Assertion (ii) of [BG, Lemma 3.6], together with its proof. Therefore the right boundary of $\mathfrak{m}^\dagger$ is an initial segment of the right boundary of the infinite map encoded by $\mathcal{Z}^\dagger$.

By [BG, Lemma 5.13], this infinite map is $\widehat{M}_{0,\infty}$ in the coupling (2.3)–(2.4). Its right boundary is the rightmost path in $M_{0,\infty}$ from x_0 to infinity, whose vertices are $x_0, x_1, \ldots$. The time-N active edge starts at $o = z_\ell$ and the subsequent lower-right boundary edges extend this path beyond o.

This proves the last assertion. Notice that on $B_N^\dagger$, Lemma 3.4 also shows that no later edge can meet the displayed finite boundary segment away from o. □

The conditioning weight (2.1) gives the following exact identities for the two possible walk paths:

$$\widehat{\mathbb{P}}\left[\widehat{\mathcal{Z}}|_{[0,N]} = \mathfrak{z}\right] = (\ell+1)3^{-N}, \tag{3.23}$$

$$\widehat{\mathbb{P}}\left[\mathcal{Z}^\dagger|_{[0,N]} = \mathfrak{z}^\dagger, B_N^\dagger\right] = (r+1)3^{-N}\frac{1}{r+1} = 3^{-N}. \tag{3.24}$$

The factor depending on r cancels. Moreover, $\ell+1 \le CU$ by (3.11), after increasing C. The original walk path is recovered uniquely from the reversed walk path by applying the same construction to its first $N-1$ increments; the final increment is $(1,-1)$ in both directions.

3.3 The marked distance after reversal

Let $v = \widehat{\lambda}_t^-$. Condition (3.9) and the upper-left boundary description in the KMSW procedure imply that v remains on the upper-left boundary of $\mathfrak{m}$. More explicitly, a direct check of the three KMSW steps shows that this boundary vertex can be removed only when the first coordinate goes below $\widehat{\mathcal{L}}(t)$.

We also have

$$\mathrm{SDP}_{\mathfrak{m}}(o,v) = \mathrm{SDP}_{\widehat{M}_{0,t}}(o,v) = a_t. \tag{3.25}$$

Indeed, Lemma 2.1, applied to the restriction to $[0,t]$, identifies its directed edge incidences with those among the same active edges in $\mathfrak{m}$. Active-edge times increase strictly along a directed path. Since $\widehat{\lambda}_t$ has initial vertex v, every edge in a directed path ending at v has active time less than t. Hence the two maps have exactly the same directed paths from o to v, which proves (3.25).

After reversing the edge orientations, v lies on the right boundary of $\mathfrak{m}^\dagger$ before its sink o, and

$$\mathrm{SDP}_{\mathfrak{m}^\dagger}(v,o) = a_t. \tag{3.26}$$

We next identify this distance with a Busemann difference. Work in the coupling (2.3)–(2.4), with $\mathcal{Z}^\dagger$ as the encoding walk of $\widehat{M}_{0,\infty}$. By Lemma 3.5, the right boundary vertices of $\mathfrak{m}^\dagger$ are the initial vertices $x_0, x_1, \dots, x_\ell$ of the right boundary of $M_{0,\infty}$. Every directed path in $M_{0,\infty}$ started from one of these vertices stays in $\widehat{M}_{0,\infty}$, by the reachability observation following (2.4). Thus SDP distances between these vertices computed in $M_{0,\infty}$ agree with those computed in $\widehat{M}_{0,\infty}$.

Furthermore, $o = (\lambda_N^\dagger)^-$. Since active times increase strictly along directed paths, every directed path which ends at o uses only edges whose active times are less than N. Later edges therefore cannot shorten the path from v to o, so

$$\mathrm{SDP}_{\mathfrak{m}^\dagger}(v,o) = \mathrm{SDP}_{M_{0,\infty}}(v,o). \tag{3.27}$$

Write $v = x_i$ and $o = x_j$. By Lemma 3.5, $0 \le i < j = \ell$ whenever $a_t > 0$. If $k > j$, then x_k belongs to the part explored after time N. Lemma 3.4, together with the fact that every directed path from v stays in $\widehat{M}_{0,\infty}$, therefore implies that every directed path from v to x_k passes through o. Hence

$$\mathrm{SDP}_{M_{0,\infty}}(v,x_k) = \mathrm{SDP}_{M_{0,\infty}}(v,o) + \mathrm{SDP}_{M_{0,\infty}}(o,x_k). \tag{3.28}$$

For the comparison below, we consider only realizations for which $a_t \ge h > 0$, so the indices above satisfy $i < j$. The right boundary of $\mathfrak{m}^\dagger$ has $\ell \le CU$ edges. The defining identity in [BG, Theorem 1.4], applied to the common targets x_k with $k > j$, together with (3.28), gives

$$a_t = \mathcal{X}(i) - \mathcal{X}(j) = -\sum_{k=i+1}^{j} Y_k. \tag{3.29}$$

In particular,

$$\max_{0\le i<j\le\lceil CU\rceil}\left(-\sum_{k=i+1}^{j}Y_k\right)\ge a_t. \tag{3.30}$$

Combining Lemma 3.2, the weight comparison (3.23)–(3.24), and (3.30) gives our key estimate.

Proposition 3.6. *There are $C>0$ and $U_0\in\mathbb{N}$ such that, for $t\in\mathbb{N}$, $U\in\mathbb{N}$ with $U\ge U_0\vee\sqrt{t}$, and $h>0$,*

$$\begin{aligned}&\widehat{\mathbb{P}}\left[a_t\ge h,\ E_t(U)\right]\\&\quad\le CU^5\mathbb{P}\left[\max_{0\le i<j\le\lceil CU\rceil}\left(-\sum_{k=i+1}^{j}Y_k\right)\ge h\right].\end{aligned} \tag{3.31}$$

Proof. Conditionally on the first t increments and on $E_t(U)$, the extension event has probability at least cU^{-4} by Lemma 3.2. Hence the left side of (3.31) is at most CU^4 times the probability of the same event together with the extension event. Sum this latter probability over all length-N walk realizations. For each realization, (3.23)–(3.24) replace its weight by at most CU times the weight of the corresponding reversed walk path together with $B_N^\dagger$. The correspondence is one-to-one, and every such walk path satisfies (3.30). Dropping the additional conditions on the reversed walk proves (3.31). □

4 Distances in the UIQBOT

We now combine the exact comparison in Proposition 3.6 with the one-sided Busemann estimate in Lemma 2.2.

Proposition 4.1. *For every $\varepsilon>0$ and every $p>0$, there is $C=C(\varepsilon,p)>0$ such that*

$$\widehat{\mathbb{P}}\left[\max_{0\le t\le m}a_t\ge m^{3/8+\varepsilon}\right]\le Cm^{-p},\qquad m\in\mathbb{N}. \tag{4.1}$$

Proof. It is enough to consider $\varepsilon<1/8$, since the estimate for any smaller positive value of ε implies the estimate for the original value. Choose $\eta\in(0,4\varepsilon/3)$ and set

$$U=\left\lceil m^{1/2+\eta}\right\rceil,\qquad h=m^{3/8+\varepsilon}. \tag{4.2}$$

Use the same value of U in Proposition 3.6 for every $t\le m$. On the event that the walk stays in the range in (2.2), each of the comparisons is valid. Moreover,

$$\frac{h^4}{U^3}=m^{4\varepsilon-3\eta+o(1)},\qquad h\ge(CU)^{3/4} \tag{4.3}$$

for all sufficiently large m. If $h>CU$, the event on the right side of (3.31) is empty. Otherwise, Lemma 2.2, followed by a union bound over $t\le m$, gives

$$\widehat{\mathbb{P}}\left[\max_{0\le t\le m}a_t\ge h,\ E_m(U)\right]\le Cm^C\exp\{-cm^{4\varepsilon-3\eta}\}. \tag{4.4}$$

By (2.2),

$$\widehat{\mathbb{P}}[E_m(U)^c]\le Cm^C\exp\{-cm^{2\eta}\}. \tag{4.5}$$

The sum of (4.4) and (4.5) is smaller than every power of m. Increasing C deals with the remaining bounded values of m. □

We record the deterministic source-reachability fact needed to pass from active initial vertices to all vertices.

Lemma 4.2. *Let $\mathfrak{Z} = (\mathfrak{L}, \mathfrak{R})$ be a finite KMSW walk started from zero and such that $\mathfrak{L}(s) \geq 0$ at every time. Let $\mathfrak{m}_s$ be the time-s map and let $o = \lambda_0^-$. Then every λ_s^- is reachable from o in $\mathfrak{m}_s$. Every vertex of $\mathfrak{m}_s$ is an endpoint of λ_q for some $q \leq s$. Consequently,*

$$\max_{z \in \mathcal{V}(\mathfrak{m}_s)} \mathrm{SDP}_{\mathfrak{m}_s}(o, z) \leq 1 + \max_{0 \leq q \leq s} \mathrm{SDP}_{\mathfrak{m}_q}(o, \lambda_q^-). \tag{4.6}$$

Proof. We prove the first assertion by induction on s. For a $(1, -1)$ step, the new active initial vertex is the preceding active terminal vertex, so it is reached by adjoining the preceding active edge. For a $(0, 1)$ step, the active initial vertex is unchanged. For a $(-1, 0)$ step, the new active initial vertex is the preceding vertex on the upper-left boundary. The alternative in the KMSW procedure which creates a new lower-left missing edge would make $\mathfrak{L}$ attain a negative value, so it cannot occur. The new active initial vertex is therefore joined to o by the directed upper-left boundary path already present in $\mathfrak{m}_{s-1}$. This completes the induction.

Each non-missing edge is the active edge at the time when it is added, and every vertex is an endpoint of a non-missing edge. Hence every vertex is an endpoint of some λ_q. If it is λ_q^-, use the first assertion. If it is λ_q^+, append the directed edge λ_q to a shortest path ending at λ_q^-. These paths remain in every later map, which proves (4.6). □

Applying this lemma to the UIQBOT encoding walk gives the following consequence without any further probabilistic estimate.

Corollary 4.3. *For every $\varepsilon > 0$ and $p > 0$, there is $C = C(\varepsilon, p) > 0$ such that*

$$\widehat{\mathbb{P}}\left[\max_{0 \leq k \leq m} \max_{z \in \mathcal{V}(\widehat{M}_{0,k})} \mathrm{SDP}_{\widehat{M}_{0,k}}(o, z) > 1 + m^{3/8+\varepsilon}\right] \leq C m^{-p}. \tag{4.7}$$

5 From the UIQBOT to the size-n KMSW cell

We now return to the finite cell $M_{0,n}$. Extend its encoding walk after time n by independent increments with the law in (1.1). For $j \in [0, n] \cap \mathbb{Z}$, put

$$\mathcal{Z}^j(s) := \mathcal{Z}(j + s) - \mathcal{Z}(j), \qquad s \geq 0, \tag{5.1}$$

and apply the KMSW procedure to $\mathcal{Z}^j$, starting from the active edge at time j. Denote the resulting map by $M_{0,\infty}^j$ and index its active edges by the corresponding times of the original walk, writing $\lambda_i^{(j)}$ for the edge with time $i \geq j$. Let Q_j be the submap induced by the vertices reachable from $o_j := (\lambda_j^{(j)})^-$. We state explicitly the part of the future-map construction which we need.

Lemma 5.1. *There are increasing KMSW times*

$$j = \sigma_j(0) < \sigma_j(1) < \sigma_j(2) < \cdots \tag{5.2}$$

such that the time-k active edge in the KMSW encoding of Q_j is $\lambda_{\sigma_j(k)}^{(j)}$. Let $Q_j[0, k]$ be the map obtained by applying the KMSW procedure to the first k steps of this encoding walk. If $\sigma_j(k) \leq n$, the correspondence

$$(\lambda_i^{(j)})^{\pm} \longmapsto \lambda_i^{\pm}, \qquad j \leq i \leq \sigma_j(k), \tag{5.3}$$

embeds $Q_j[0, k]$ as a directed submap of $M_{0,n}$.

Proof. Apply [BG, Lemma 5.13] to the shifted exploration $M^j_{0,\infty}$. That lemma is a pathwise statement: the KMSW encoding of the induced reachable submap is obtained by retaining, in chronological order, precisely the active edges which belong to that submap, with their endpoint incidences unchanged. Thus the non-missing edges of $Q_j[0,k]$ are

$$\lambda^{(j)}_{\sigma_j(0)}, \lambda^{(j)}_{\sigma_j(1)}, \dots, \lambda^{(j)}_{\sigma_j(k)}.$$

Every non-missing edge of a KMSW map is the active edge at its time of addition. For the finite interval $[j, \sigma_j(k)]$, Lemma 2.1 identifies the endpoints of every $\lambda^{(j)}_i$ with those of the corresponding λ_i in $M_{0,n}$, and it preserves both equality and inequality of endpoints. The correspondence (5.3) is therefore an injective directed graph homomorphism on all edges of $Q_j[0,k]$. This is the asserted embedding. □

By [BG, Proposition 5.1 and Lemmas 5.13–5.14],

$$((Q_j[0,k])_{k\geq 0}, o_j) \stackrel{d}{=} ((\widehat{M}_{0,k})_{k\geq 0}, o) \tag{5.4}$$

for each deterministic j.

Fix distinct vertices $x, y \in \mathcal{V}(M_{0,n})$ with $\mathrm{SDP}_{M_{0,n}}(x,y) < \infty$, and choose a shortest directed path $P = (e_1, \dots, e_d)$ from x to y. Let j be the active time of e_1. By Lemma 2.1, the active times of $e_1, \dots, e_d$ increase strictly and all belong to $[j,n]$. The last part of that lemma transports P, with all of its directed incidences, to the KMSW exploration of (5.1). In particular, this transported path starts from o_j, so each of its vertices and edges belongs to Q_j.

Let k be the index in (5.2) of the last edge e_d. Then $\sigma_j(k) \leq n$, so $k \leq n+1$. All the preceding edges of P occur among the first k active edges of Q_j, since their KMSW times are smaller. Hence P is a directed path from $o_j = x$ to y in $Q_j[0,k]$. By Lemma 5.1,

$$Q_j[0,k] \subset M_{0,n}. \tag{5.5}$$

A shortest path from o_j to y in $Q_j[0,k]$ is therefore also a path in $M_{0,n}$. Consequently,

$$\mathrm{SDP}_{M_{0,n}}(x,y) \leq \mathrm{SDP}_{Q_j[0,k]}(o_j, y) \leq \max_{z \in \mathcal{V}(Q_j[0,k])} \mathrm{SDP}_{Q_j[0,k]}(o_j, z). \tag{5.6}$$

Equations (5.4) and (5.6) reduce all ordered pairs to the estimate of Corollary 4.3. No independence between the maps Q_j is needed. The case $x = y$ has SDP distance zero and needs no separate estimate.

Proof of Theorem 1.1. Apply Corollary 4.3 with $m = n+1$ and with $\varepsilon/2$ in place of ε. Take a union bound over $j \in [0,n] \cap \mathbb{Z}$ in (5.6). For every $p > 0$, after first using the corollary with a sufficiently large power, this gives

$$\mathbb{P}\left[D_n > 1 + (n+1)^{3/8+\varepsilon/2}\right] \leq Cn^{-p}. \tag{5.7}$$

For all sufficiently large n, $1 + (n+1)^{3/8+\varepsilon/2} < n^{3/8+\varepsilon}$. Enlarging C to include the remaining values of n proves (1.3). □

6 The median exponent

Proof of Corollary 1.2. Fix $\zeta \in (0, 1/2)$. By the third assertion of [BG, Theorem 1.13], there is $c_\zeta > 0$, independent of n, such that

$$\mathbb{P}\left[D_n \geq c_\zeta n^{3/8}\right] \geq 1 - \zeta, \qquad n \in \mathbb{N}. \tag{6.1}$$

Since $1 - \zeta > 1/2$, any median satisfies $m_n \geq c_\zeta n^{3/8}$.

On the other hand, Theorem 1.1 implies that, for every $\varepsilon > 0$ and all sufficiently large n,

$$\mathbb{P}[D_n \geq n^{3/8+\varepsilon}] < \frac{1}{2}. \tag{6.2}$$

Hence $m_n \leq n^{3/8+\varepsilon}$ for all such n. It follows that

$$\frac{3}{8} + \frac{\log c_\zeta}{\log n} \leq \frac{\log m_n}{\log n} \leq \frac{3}{8} + \varepsilon. \tag{6.3}$$

First send $n \to \infty$, then send $\varepsilon \to 0$, to obtain (1.5). □

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